Visvam Rajesh

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CS student building ML-driven autonomous and high-performance systems. Looking for internships.

Skills

Languages	Python, C++, Rust, Java, Rust
Robotics & Systems	ROS 2, Linux, Docker
ML/AI	PyTorch, scikit-learn, NumPy
Tools	AWS (EC2), Git, CI/CD

Education

2025 - 2028 **Carnegie Mellon University** BS Computer Science & Robotics

Coursework: Data Structures, Algorithms, Imperative Computation, Theoretical Computer Science, Discrete Math

Work Experience

Foam Robotics Lab, CMU Robotics Institute Feb 2026 - Present

Researcher, Motion Planning Lead Architecting the first simulation stack for the novel DexKit surgical arm (C++/Python, ROS2), building URDF-based MuJoCo models for deployment in UPMC operating rooms. Implementing constraint-aware motion planning (MoveIt + Risk-DTRRT) with collision avoidance and limited-DOF kinematics for safe tissue retractor manipulation.

Carnegie Mellon Racing Sept 2025 - Present

Driverless Path-Planning Developer Designed a real-time (20Hz) custom trajectory planner computing cone-derived midlines for autonomous racecar navigation (C++/ROS2). Improved simulation consistency in Unreal Engine 4 via curve-fitting cone interpolation, enabling sub-60s autonomous lap completion.

ScottyLabs Sept 2025 - Present

Software Engineer Built Rust-based scheduling engine optimizing course sequences by major constraints and minimizing walking distance (MongoDB backend). Reduced load times from 1200ms to 200ms; maintained reliability with 5k+ concurrent users during peak registration.

Visra Solutions Dec 2023 - Aug 2025

Software Engineer Intern Rapidly built and deployed full-stack web applications (React/Gatsby/Next.js, Express) on AWS EC2. Designed modular, performance-oriented UI components and internal tooling for maintainable front-end systems.

The Daleks, FIRST Robotics Competition Team 3637 Sept 2021 - Jun 2025

President (June 2024 – June 2025), **Programming Captain** (May 2023 - June 2024)

Led 60+ member multidisciplinary robotics team; restructured executive board to establish dedicated VP of Software, improving engineering resource allocation. Built ROS-based LiDAR + vision pipeline, simulation, and data-logging system; increased autonomous scoring from 1–2 to 4–5 game pieces per match.

Projects

Poké-ViT (Link) Jan 2026

Implemented Vision Transformer from scratch in PyTorch (CUDA 13, RTX 4060), training on 40k images across 1000 classes. Achieved 56% top-1 accuracy with augmentation (random flips, normalization, 256×256 inputs). Engineered end-to-end data pipeline with custom web scraping and preprocessing for large-scale classification.

Password Manager (Link) July 2025

Built cross-platform encrypted password manager in Python (GPLv3) with AES-based Fernet encryption, salted hashing, and OS keychain session caching. Designed single-file secure storage with auto-locking sessions, fuzzy search, CLI scripting interface, and Textual-based TUI.

Publications

Rajesh, V. & Wu, C. An Extension of Pathfinding Algorithms for Randomly Determined Speeds. IEEE International Performance, Computing, and Communications Conference, 2024. (Link)